

On the Categorical Mechanics of Real-Time Event Selection in Bounded Phase Space: Composition-Inflation, Temporal Programming, and S-Entropy Trajectory Completion in LHC Trigger Systems

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May 31, 2026

Abstract

We develop a unified mathematical framework for real-time event selection in Large Hadron Collider (LHC) trigger systems, grounded in three independently motivated but mutually reinforcing structures: (i) the *composition-inflation mechanism*, which establishes that a trigger system with d independent detector subsystems observing n bunch crossings can discriminate $T(n, d) = d(d+1)^{n-1}$ categorically distinguishable event patterns—an exponential, not linear, state count; (ii) *temporal programming*, a paradigm in which the sole runtime datum is the timing deviation $\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k)$ of the k -th detector signal from the LHC bunch clock, and in which the trigger’s cell registry constitutes a formally verified, structurally incorruptible specification with no injection surface; and (iii) *S-entropy backward trajectory completion*, in which the trigger’s classification task is identified with navigation from a known endpoint (the detected event) to its root (the originating physics process) in a ternary refinement hierarchy, achieving $\mathcal{O}(\log_3 N)$ complexity via virtual sub-state decompositions.

We derive the *Planck depth* $n_P = 60$ for the LHC bunch clock ($\nu_B = 40.079$ MHz, $d = 3$): after 60 bunch crossings ($1.496 \mu\text{s}$), the trigger’s categorical pattern count $T(60, 3) = 3 \cdot 4^{59} \approx 7.3 \times 10^{35}$ exceeds the ratio $\tau_B/t_P = 4.63 \times 10^{35}$ of the bunch-crossing period to the Planck time. The current L1 trigger latency of 100 bunch crossings operates at depth $n/n_P \approx 1.7$, placing it entirely within the super-Planck categorical resolution regime.

We establish a formal identification between *virtual particles* in quantum field theory and *virtual sub-states* in the S-entropy framework: off-shell propagation ($q^2 \neq m^2$) corresponds precisely to sub-coordinate decompositions outside the physical unit cube ($\mathbf{S} \notin [0, 1]^3$), and the five canonical properties of virtual particles—enabling transitions, zero net work, path opacity, mediating catalytic power, and necessity for exponential

speedup—are theorems of the categorical framework, not physical assumptions. The Ward–Takahashi identities of gauge field theory are identified with the Local-Global Decoupling Theorem, and S-matrix observability is identified with Path Opacity.

We derive all seven fundamental constants in categorical angular units: $c = 2\pi$ rad/tick, $\hbar = E_{\text{tick}}/(2\pi)$, $k_B = E_{\text{tick}}/\ln(d+1)$, $m_0 = E_{\text{tick}}\nu_0/(4\pi^2)$, with the gravitational constant G remaining the unique irreducible coupling not reducible to tick counting and 2π geometry. The HL-LHC upgrade to $d = 4$ independent subsystems reduces the Planck depth from 60 to 52, recovering eight bunch crossings of categorical resolution headroom.

Keywords: LHC trigger, composition-inflation, temporal programming, S-entropy, backward trajectory completion, virtual sub-states, virtual particles, Planck depth, Ward identity, S-matrix, categorical mechanics, bunch crossing, coincidence detection

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1 Introduction

1.1 The Event Selection Problem

The Large Hadron Collider at CERN produces proton–proton collisions at a rate of 40.079 MHz (Brüning et al., 2004; Evans and Bryant, 2008). Each collision, if fully recorded, generates approximately 1–2 MB of raw detector data (ATLAS Collaboration, 2008; CMS Collaboration, 2008). The aggregate data rate—approaching 80 TB per second—cannot be recorded by any existing storage technology. The trigger system must reduce this rate to approximately 1 kHz for permanent storage, rejecting 99.997% of all events in real time (ATLAS Collaboration, 1998, 2020).

This event selection problem has three structural properties that distinguish it from ordinary digital filtering:

1. **Real-time irreversibility.** The decision to discard an event is permanent: the detector data is overwritten in the analog pipeline before the next bunch crossing arrives. No second chance is available (As-samagan et al., 2003; Cittolin et al., 2002).

2. **Timing primacy.** The sole physically guaranteed distinction between a signal event and background is the timing of detector signals relative to the LHC bunch clock. Content-bearing information (particle identities, masses, charges) is inferred from timing patterns, not transmitted as content (Leo, 1994).
3. **Exponential state space.** The number of distinct event topologies accessible at each bunch crossing is not n (the number of crossings observed) but grows exponentially with n —a fact not captured by any existing formal framework for trigger design.

The present paper establishes a rigorous mathematical framework that makes these three properties precise, derives their consequences, and produces new design principles for the LHC and its HL-LHC upgrade.

1.2 Overview of Main Results

The central results are organised as follows.

Composition-Inflation (Section 3). We prove that a trigger system with d independent detector sub-systems, observing n bunch crossings, can discriminate

$$T(n, d) = d \cdot (d + 1)^{n-1} \quad (1)$$

categorically distinguishable event patterns. This exponential growth arises from two independent multiplications: 2^{n-1} ordered compositions of n bunch crossings (the temporal structure), and d^k dimensional labelings of k -part compositions (the spatial structure). The naïve linear count of n states per channel misses the combinatorial richness by a factor $d(d + 1)^{n-1}/n$, which exceeds 10^{28} for $n = 56$ at $d = 3$.

Planck Depth for the LHC (Section 4). We define the *Planck depth* n_P as the minimum number of bunch crossings at which $T(n_P, d)$ exceeds the ratio τ_B/t_P of the bunch-crossing period to the Planck time. For the LHC at $d = 3$, $n_P = 60$ ($1.496 \mu\text{s}$). The current L1 trigger latency of 100 bunch crossings operates at depth 1.7 times the Planck depth.

Temporal Programming (Section 5). We formalise the trigger as a *Tempus program*: a compiled cell registry over the timing-deviation space $\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k)$. The program has no content parser and no injection surface, satisfying a *structural incorruptibility* theorem: no signal whose timing does not originate from the LHC bunch crossing can satisfy any trigger cell condition.

Backward Trajectory Completion (Section 6). We identify the trigger decision problem as backward navigation in a ternary refinement hierarchy: the end-point (detector hit pattern) is known, and the root (originating physics process) must be found. Using the S-entropy embedding of the hierarchy into $[0, 1]^3$ with the product Fisher metric, backward navigation achieves $\mathcal{O}(\log_3 N)$ complexity. Restricting to physical intermediate states collapses to $\Theta(N)$, proving that *virtual sub-states are computationally necessary*.

Virtual Particles as Virtual Sub-States (Section 7). We prove a formal identification theorem:

virtual particles in quantum field theory are virtual sub-states in the S-entropy framework. The off-shell condition ($q^2 \neq m^2 c^4$) corresponds to lying outside $[0, 1]^3$; the Feynman propagator corresponds to the mean-recovery constraint; gauge invariance corresponds to Local-Global Decoupling; and S-matrix observability corresponds to Path Opacity.

Constant Reduction (Section 8). All seven fundamental constants reduce to tick counts, 2π geometry, and tick energy in the categorical angular framework, with G as the unique irreducible coupling. This implies the Planck time is not a fundamental resolution limit but a composition depth: $t_P \leftrightarrow n_P \approx 56\text{--}72$ for any physical oscillator.

2 Mathematical Foundations: Bounded Phase Space

2.1 The Bounded Phase Space Axiom

All physical systems in this framework occupy bounded regions of phase space. This is not an assumption but a physical necessity: no system possesses infinite energy or infinite momentum.

Axiom 1 (Bounded Phase Space). Every physical system occupies a bounded region $\Omega \subset \mathcal{P}$ of phase space with finite Liouville measure: $\mu(\Omega) < \infty$.

Theorem 2.1 (Partition Existence [Arnold, 1989](#)). For any bounded region Ω with $\mu(\Omega) < \infty$, there exists a finite partition $\{\Omega_i\}_{i=1}^N$ with $N < \infty$ such that $\Omega = \bigsqcup_{i=1}^N \Omega_i$.

Proof. By compactness of bounded sets in phase space (Heine–Borel theorem), any open cover of Ω admits a finite subcover. The partition follows from refinement of such a cover; finiteness of N follows from $\mu(\Omega) < \infty$ requiring each Ω_i to have positive measure. $\square$ $\square$

2.2 Oscillatory Necessity and Action-Angle Variables

Theorem 2.2 (Poincaré Recurrence [Poincaré, 1890](#)). Let $\Sigma_E = H^{-1}(E)$ be a compact energy surface with $\mu(\Sigma_E) < \infty$. For any measurable set $A \subset \Sigma_E$ with $\mu(A) > 0$, almost every trajectory starting in A returns to A infinitely often.

Corollary 2.3 (Oscillatory Necessity). Every compact Hamiltonian system without fixed points is oscillatory: no coordinate can undergo monotonic escape, and every trajectory returns to any neighbourhood of its initial condition.

Proof. Monotonic escape contradicts Theorem 2.2 on a compact energy surface. Absence of fixed points (guaranteed for systems at $T > 0$ by the Third Law) forces the trajectory to oscillate. $\square$ $\square$

For integrable systems, the Arnold–Liouville theorem ([Arnold, 1989](#)) guarantees the existence of action-angle variables (J_i, θ_i) with quasi-periodic motion $\dot{\theta}_i = \omega_i(J)$. Closure of orbits requires $n_i \omega_i = n_j \omega_j$ for positive integers n_i ; the tuple (n_1, n_2, n_3) in three dimensions labels the *partition state*.

2.3 Partition Coordinates and Capacity

Definition 2.4 (Partition State). A *partition state* of a bounded three-dimensional Hamiltonian system is an equivalence class of phase-space points sharing the same closure label (n, ℓ, m, s) where: $n \geq 1$ (principal level), $0 \leq \ell < n$ (angular sublevel), $-\ell \leq m \leq \ell$ (azimuthal projection), $s = \pm \frac{1}{2}$ (rotational parity). The set of all partition states is $\mathcal{P}$.

Theorem 2.5 (Partition Capacity). The number of distinct partition states at principal level n is $C(n) = 2n^2$.

Proof. At level n , ℓ ranges over $\{0, 1, \dots, n-1\}$; for each ℓ , m ranges over $2\ell+1$ values; and s contributes a factor of 2:

$$C(n) = 2 \sum_{\ell=0}^{n-1} (2\ell+1) = 2 \left(n + 2 \cdot \frac{n(n-1)}{2} \right) = 2n^2. \quad \square$$

$\square$

The cumulative count is $N_{\text{state}}(n) = \frac{n(n+1)(2n+1)}{3}$, growing cubically.

3 Composition-Inflation

3.1 Integer Compositions

Definition 3.1 (Composition). A *composition* of a positive integer n is an ordered sequence $(c_1, c_2, \dots, c_k)$ of positive integers summing to n . Two compositions differing in order are distinct.

Theorem 3.2 (Composition Count [Andrews, 1976](#)). The total number of compositions of n is 2^{n-1} .

Proof. Place n objects in a row; there are $n-1$ gaps between consecutive objects. A composition corresponds to a binary choice (place/do not place a divider) at each gap: 2^{n-1} configurations in bijection with compositions. $\square$ $\square$

The number of compositions into exactly k parts is $\binom{n-1}{k-1}$ (choose $k-1$ of the $n-1$ gaps for dividers).

3.2 Dimensional Labeling

The S-entropy space is the three-dimensional unit cube $[0, 1]^3$ with coordinates (S_k, S_t, S_e) representing knowledge, temporal, and categorical entropy dimensions, respectively.

Definition 3.3 (Labeled Composition). A *labeled composition* of n in d dimensions is a pair $(\mathbf{C}, \mathbf{L})$ where $\mathbf{C} = (c_1, \dots, c_k)$ is a composition and $\mathbf{L} = (\ell_1, \dots, \ell_k)$ with $\ell_i \in \{1, \dots, d\}$ specifies the entropy dimension being refined in the i -th part.

Theorem 3.4 (Composition-Inflation Formula). *The total number of labeled compositions of n in d dimensions is*

$$T(n, d) = d \cdot (d+1)^{n-1}. \quad (2)$$

Proof. Summing over all part counts:

$$T(n, d) = \sum_{k=1}^n \binom{n-1}{k-1} d^k = d \sum_{j=0}^{n-1} \binom{n-1}{j} d^j = d(1+d)^{n-1} \quad (3)$$

where the last step is the binomial theorem with $x = d$, $m = n-1$. $\square$

Corollary 3.5 (S-Entropy Space). *For $d = 3$: $T(n, 3) = 3 \cdot 4^{n-1}$. After $n = 56$ cycles, $T = 3 \cdot 4^{55} \approx 3.8 \times 10^{33}$, exceeding 10^{33} distinguishable trajectories.*

Proposition 3.6 (Growth Properties). *The labeled composition count satisfies:*

- (i) $T(1, d) = d$;
- (ii) $T(n+1, d) = (d+1) \cdot T(n, d)$ (each tick multiplies states by $d+1$);
- (iii) $\log T(n, d) = \log d + (n-1) \log(d+1)$;
- (iv) $T(n, d)/n \rightarrow \infty$ as $n \rightarrow \infty$ (superlinear growth).

Proof. Direct from Theorem 3.4. $\square$

3.3 Angular Resolution and the Planck Constraint

Definition 3.7 (Angular Resolution). For $T(n, d)$ distinguishable trajectories distributed over the phase circle $[0, 2\pi)$:

$$\Delta\theta(n, d) = \frac{2\pi}{T(n, d)} = \frac{2\pi}{d(d+1)^{n-1}}.$$

Theorem 3.8 (No Planck Limit on Angular Resolution). *The angular resolution $\Delta\theta(n, d)$ is dimensionless (units of radians, not seconds) and therefore not subject to the Planck time constraint. For any $\epsilon > 0$, there exists $n_0 = 1 + \lceil \log_{d+1}(2\pi/(d\epsilon)) \rceil$ such that $\Delta\theta < \epsilon$ for all $n \geq n_0$.*

Proof. The Planck time $t_P = \sqrt{\hbar G/c^5}$ has dimension $[t_P] = \text{s}$. The Heisenberg bound $\Delta E \cdot \Delta t \geq \hbar/2$ constrains energy-time products; both factors carry physical dimensions. The angular resolution $\Delta\theta = 2\pi/T(n, d)$ is the ratio of two dimensionless quantities and carries no time dimension: $[\Delta\theta] = \text{rad} = 1$. Neither the Heisenberg argument nor the gravitational collapse argument applies to a dimensionless ratio. The lower bound on n_0 follows from $d(d+1)^{n-1} > 2\pi/\epsilon$, solved for n . $\square$

4 The Planck Depth for LHC Trigger Systems

Definition 4.1 (Planck Depth). The *Planck depth* n_P for a reference oscillator with period τ_{osc} in d -dimensional S-entropy space is the minimum cycle count for which

$$T(n_P, d) \geq \frac{\tau_{\text{osc}}}{t_P}.$$

Theorem 4.2 (Planck Depth Formula).

$$n_P = 1 + \left\lceil \log_{d+1} \left(\frac{\tau_{\text{osc}}}{d \cdot t_P} \right) \right\rceil.$$

Proof. Solving $d(d+1)^{n_P-1} \geq \tau_{\text{osc}}/t_P$ for n_P yields $n_P - 1 \geq \log_{d+1}(\tau_{\text{osc}}/(d t_P))$; the minimum integer satisfying this is the given expression. $\square$

Theorem 4.3 (LHC Planck Depth). *For the LHC bunch clock ($\nu_B = 40.079$ MHz, $\tau_B = 24.951$ ns) with $d = 3$ independent detector subsystems, the Planck depth is $n_P = 60$, reached in physical time $t_{60} = 60/\nu_B = 1.496$ μs .*

Proof.

$$\frac{\tau_B}{t_P} = \frac{24.951 \times 10^{-9}}{5.391 \times 10^{-44}} = 4.628 \times 10^{35}. \quad (4)$$

Applying Theorem 4.2 with $d = 3$:

$$\begin{aligned} n_P &= 1 + \left\lceil \log_4 \left(\frac{4.628 \times 10^{35}}{3} \right) \right\rceil = 1 + \lceil \log_4(1.543 \times 10^{35}) \rceil \\ &= 1 + \left\lceil \frac{\ln(1.543 \times 10^{35})}{\ln 4} \right\rceil = 1 + \left\lceil \frac{81.10}{1.386} \right\rceil = 1 + \lceil 58.51 \rceil = 60. \end{aligned} \quad (5)$$

Verification: $T(60, 3) = 3 \cdot 4^{59} = 3 \cdot 3.076 \times 10^{35} = 9.23 \times 10^{35} > 4.628 \times 10^{35}$. $\square$

Corollary 4.4 (Universality of Planck Depth). *For any physically realizable oscillator (frequency in $[\nu_{\min}, \nu_{\max}]$ with $t_P^{-1} > \nu_{\max}$ and $\nu_{\min} > 0$), the Planck depth at $d = 3$ satisfies $n_P \in [36, 73]$. The variation spans only 37 ticks despite frequencies varying over 14 orders of magnitude, because n_P depends logarithmically on ν .*

Table 1: Planck depth for representative LHC-relevant oscillators ($d = 3$).

Oscillator	ν	τ/t_P	n_P
LHC bunch clock	40.079 MHz	4.63×10^{35}	60
Cs-133 hyperfine	9.193 GHz	2.02×10^{33}	56
H maser	1.420 GHz	1.31×10^{34}	57
Sr optical clock	429 THz	4.32×10^{28}	48
HL-LHC ($d=4$)	40.079 MHz	4.63×10^{35}	52

The HL-LHC row demonstrates the dimensional advantage: adding a fourth independent detector subsystem (precision timing, $d : 3 \rightarrow 4$) reduces n_P from 60 to 52, recovering 8 bunch crossings (200 ns) of categorical resolution headroom.

Theorem 4.5 (HL-LHC Dimension Advantage). *For the LHC bunch clock with $d = 4$ (adding the precision timing detector planned for HL-LHC):*

$$n_P(d = 4) = 1 + \left\lceil \log_5 \left(\frac{4.628 \times 10^{35}}{4} \right) \right\rceil = 52.$$

The dimensional upgrade saves $n_P(3) - n_P(4) = 8$ bunch crossings at fixed oscillator frequency.

Proof. $\log_5(4.628 \times 10^{35}/4) = \log_5(1.157 \times 10^{35}) = \ln(1.157 \times 10^{35})/\ln 5 = 80.5/1.609 = 50.03$; $n_P = 1 + 51 = 52$. $\square$ $\square$

5 Temporal Programming and Structural Incorruptibility

5.1 The Timing Deviation as Sole Datum

Definition 5.1 (ΔP -Datum). For a reference oscillator with period T_{ref} and a detector signal arriving at time $t_{\text{rec}}(k)$, the *timing deviation* of the k -th signal is

$$\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k).$$

In the LHC context, $T_{\text{ref}}(k)$ is the k -th bunch crossing time from the 40 MHz bunch clock, and $t_{\text{rec}}(k)$ is the recorded hit time in any detector channel.

Definition 5.2 (Timing Cell). A *timing cell* C_i is a Borel-measurable subset of the ΔP -space $\mathbb{R}$, with associated action a_i . The collection $\{C_i\}$ forms the *cell registry* Γ , which must partition the relevant range of ΔP values.

Definition 5.3 (Temporal Trigger Program). A *temporal trigger program* is a tuple $\langle \sigma, \Gamma, \tau, \phi \rangle$ where $\sigma \in \{\text{IDLE}, \text{COMPILE}, \text{EXECUTE}\}$ is the phase machine state, Γ is the cell registry, τ is the timing reference channel specification, and ϕ is the action dispatch function mapping cell membership to trigger decisions.

5.2 Structural Incorruptibility

Theorem 5.4 (Structural Incorruptibility of Timing Triggers). *A trigger program that conditions entirely on $\Delta P(k)$ values—with no content parser applied to signal amplitude, charge, or identity—has no injection surface: any signal whose timing $\Delta P(k)$ does not originate from the LHC bunch crossing will fail to satisfy any physically calibrated cell condition.*

Proof. The cell registry partitions ΔP -space. A signal from a real collision produced at the bunch crossing time $T_{\text{ref}}(k)$ satisfies $|\Delta P(k)| < \delta$ for the bunch-crossing timing jitter δ (typically $\lesssim 1$ ns for calibrated detectors (ATLAS Collaboration, 2020)). A signal from noise, cosmic rays, or beam halo arrives with a $\Delta P(k)$ value drawn from a distribution with mean offset from $T_{\text{ref}}(k)$. Since the cell conditions C_i are defined by ΔP membership alone, and the cell boundaries are calibrated to the

collision timing distribution, there is no content layer (no message, no encoding, no data packet) on which an attacker can operate. The only way to satisfy a cell condition is to produce a signal with the correct timing relative to T_{ref} , which requires physical presence at the interaction point. $\square$ $\square$

Remark 5.5. This theorem identifies the timing trigger as the particle physics analogue of a structurally incorruptible communication channel: the “message” (the physics process) is observed through timing structure, not transmitted as content. Compare with the Phase-Locked Messaging framework (Landauer, 1961), where security emerges from topological structure rather than cryptographic secrets.

5.3 The Trigger as a Tempus Program

We write the basic di-muon trigger as a temporal program in outline:

1. **Reference:** The LHC bunch clock at 40.079 MHz provides $T_{\text{ref}}(k)$.
2. **Cells:** $C_{\text{prompt}} = \{|\Delta P| < 12.5 \text{ ns}\}$; $C_{\text{background}} = \{12.5 < |\Delta P| < 100 \text{ ns}\}$; $C_{\text{noise}} = \{|\Delta P| > 100 \text{ ns}\}$.
3. **Composition:** Require two prompt signals in the muon subsystem within one bunch crossing interval (coincidence).
4. **Dispatch:** On coincident pair in C_{prompt} , fire $a = \text{ACCEPT}$; otherwise $a = \text{DISCARD}$.

The composition rule “two prompt signals within one crossing” is the physical trigger condition formalised as a labeled composition of depth 2 in the muon cell registry.

6 The Trigger as Backward Trajectory Completion

6.1 The Trigger Decision Problem as Trajectory Completion

The fundamental task of the trigger is: given a pattern of detector hits (the *endpoint*), determine which physics process produced it (the *root*). This is precisely backward trajectory completion.

Definition 6.1 (Ternary Refinement Hierarchy). A *ternary refinement hierarchy* of depth k is a partition structure $(\Omega, \{P_j\}_{j=0}^k, \pi)$ where $|P_0| = 1$, $|P_{j+1}| = 3|P_j|$, and each block of P_j refines into exactly three children in P_{j+1} . The total leaf count is $N = 3^k$.

In the trigger context: depth k corresponds to k levels of detector subsystem hierarchy (L1 calorimeter primitives, L1 muon system, HLT tracking); each level refines the event classification into three broad categories (hadronic background, semi-leptonic candidates, fully leptonic/rare signals).

Definition 6.2 (S-Entropy Embedding). The *S-entropy embedding* of a ternary refinement hierarchy of depth k is the map $\iota_k: P_k \rightarrow [0, 1]^3$ sending each leaf $p \in P_k$ to the triple $(S_k(p), S_t(p), S_e(p))$ given by the base-3 expansion of the leaf's three coordinate strings:

$$S_k(p) = \sum_{j=1}^k \frac{a_j^k(p)}{3^j}, \quad S_t(p) = \sum_{j=1}^k \frac{a_j^t(p)}{3^j}, \quad S_e(p) = \sum_{j=1}^k \frac{a_j^e(p)}{3^j},$$

where $a_j^c(p) \in \{0, 1, 2\}$ are the address digits of p at level j .

Theorem 6.3 (Trigger State Inflation). *A trigger system with d independent detector subsystems, embedded via ι_k and observing $n = k$ levels of hierarchy, can discriminate $T(n, d) = d(d+1)^{n-1}$ distinct event categories.*

Proof. Apply Theorem 3.4: the $n = k$ levels of the hierarchy correspond to n oscillation ticks; the d detector subsystems correspond to d entropy dimensions. Each labeled composition of n ticks in d dimensions is a distinct trajectory through the hierarchy, and distinct trajectories map to distinct leaves under the embedding ι_k by the base-3 expansion (different address strings). $\square$

6.2 Backward Navigation Complexity

Theorem 6.4 (Backward Navigation Complexity). *Backward navigation from any leaf $p_k \in P_k$ to the root p_0 via the S-entropy embedding visits exactly $\log_3 N = k$ intermediate states, where $N = 3^k$, achievable in $\mathcal{O}(\log_3 N)$ steps.*

Proof. Each backward step from p_j to p_{j-1} drops the last digit of the base-3 expansion of $\iota_j(p_j)$, an $\mathcal{O}(1)$ operation. Starting at p_k and ending at p_0 requires $k = \log_3 N$ steps. $\square$

Theorem 6.5 (Collapse Without Virtual States). *Backward trajectory completion restricted to intermediate states in $[0, 1]^3$ (physical sub-states only) has worst-case complexity $\Theta(N)$.*

Proof. Without sub-coordinates outside $[0, 1]^3$, the backward step from p_j requires enumerating the children of each candidate parent at level $j-1$. At level j there are 3^j candidate parent blocks, each with three children. Summing over all levels: $\sum_{j=0}^k 3^j = (3^{k+1} - 1)/2 = \Theta(N)$. $\square$

Corollary 6.6. *Virtual intermediate states are not optional for efficient trigger computation: restricting to physically realised intermediate states increases the computational cost from $\mathcal{O}(\log N)$ to $\Theta(N)$, making real-time event selection at 40 MHz impossible.*

6.3 Virtual Sub-States in the Trigger

Definition 6.7 (Virtual Sub-State). A *virtual sub-state* is an intermediate decomposition $(s_1, s_2, s_3) \in \mathbb{R}^3$ with mean $\bar{s} = s \in [0, 1]$ but at least one component $s_i \notin [0, 1]$.

In the trigger context, virtual sub-states arise naturally: when the HLT reconstructs a track, it uses the Kalman filter equations, which produce intermediate state estimates that may transiently exceed physical bounds before converging. These are not artifacts—they are the mechanism by which the filter achieves its $\mathcal{O}(N_{\text{det}})$ rather than $\mathcal{O}(N_{\text{det}}^2)$ complexity.

7 Virtual Particles as Virtual Sub-States

7.1 Review of Virtual Particles in QFT

In quantum field theory, a *virtual particle* is an internal line in a Feynman diagram (Feynman, 1949; Peskin and Schroeder, 1995). A real (external) particle with mass m satisfies the on-shell condition $q^2 = q^\mu q_\mu = m^2 c^4$ in the metric convention $(+, -, -, -)$. A virtual particle has 4-momentum q^μ satisfying $q^2 \neq m^2 c^4$; it is *off-shell*. Its contribution to an amplitude is given by the Feynman propagator:

$$D_F(q) = \frac{i}{q^2 - m^2 c^4 + i\epsilon}. \quad (6)$$

The propagator diverges as $q^2 \rightarrow m^2$, which is the on-shell limit. Virtual particles cannot be directly observed—only the external (on-shell) particles appear as asymptotic states in the S-matrix (Dyson, 1949; Heisenberg, 1943).

7.2 The Formal Identification Theorem

We now establish the formal correspondence between virtual particles in QFT and virtual sub-states in the S-entropy framework.

Theorem 7.1 (Virtual Particle Identification). *Virtual particles in quantum field theory are formal virtual sub-states in the S-entropy framework. Specifically, the following five correspondences hold:*

- (i) **Off-shell** $\leftrightarrow$ **Outside** $[0, 1]^3$: The on-shell condition $q^2 = m^2 c^4$ corresponds to the physical constraint $\mathbf{S} \in [0, 1]^3$. Off-shell states ($q^2 \neq m^2 c^4$) correspond to virtual sub-states ($\mathbf{S} \notin [0, 1]^3$).
- (ii) **Propagator** $\leftrightarrow$ **Mean-recovery**: The Feynman propagator $D_F(q) = i/(q^2 - m^2 + i\epsilon)$ encodes the constraint that virtual momentum sums to give the physical external momentum. This is the mean-recovery constraint: $(s_1 + s_2 + s_3)/3 = s \in [0, 1]$.
- (iii) **Gauge invariance** $\leftrightarrow$ **Local-Global Decoupling**: Gauge invariance of the S-matrix means the physical amplitude is independent of the gauge choice (the local S-values of virtual contributions). This is precisely the Local-Global Decoupling Theorem: the global S-value is independent of the local S-values of subtasks.
- (iv) **S-matrix observability** $\leftrightarrow$ **Path Opacity**: In QFT, only external-state amplitudes are observable;

the virtual intermediate states are unobservable. This is Path Opacity: two backward trajectories sharing an endpoint are observationally indistinguishable.

- (v) **Unitarity $\leftrightarrow$ Cascade Power:** *The unitarity of the S-matrix ($SS^\dagger = 1$) corresponds to $\sum_i \kappa_i$ diverging, i.e. the cascade of virtual contributions saturates to complete problem resolution.*

Proof. We verify each correspondence.

(i): The physical state space of relativistic particles is the mass hyperboloid $\mathcal{M}_m = \{q^\mu : q^2 = m^2\}$ in Minkowski space. The physical state space in the S-entropy framework is the unit cube $[0, 1]^3$. The mapping is: on-shell $\leftrightarrow$ inside the physical domain; off-shell $\leftrightarrow$ outside. This is a domain identification, not a numerical equality.

(ii): The Feynman propagator at a vertex enforces 4-momentum conservation: if an external pair (p_1, p_2) produces a virtual particle of momentum $q = p_1 + p_2$, the propagator weight $i/(q^2 - m^2 + i\epsilon)$ encodes the constraint that the virtual momentum sums to give the physical input momentum. Analogously, the mean-recovery constraint $(s_1 + s_2 + s_3)/3 = s$ encodes that virtual sub-components sum (mean) to give the physical global coordinate. Both are linear constraints on intermediate values.

(iii): In gauge theory, the S-matrix satisfies $\mathcal{M} = \mathcal{M}(\text{physical states})$ independent of the gauge parameter ξ in the gauge-fixing term $-(\partial^\mu A_\mu)^2/(2\xi)$. The Local-Global Decoupling Theorem states: for any global S-value $\mathcal{S}^*$ and any assignment of local S-values to subtasks, there exists an expression realising both simultaneously. In the gauge theory interpretation: for any physical amplitude $\mathcal{M}$ (global value), the local amplitudes of individual Feynman diagrams (local S-values) can take any gauge-dependent value without changing $\mathcal{M}$ (Peskin and Schroeder, 1995). This is a statement about freedom of virtual contribution assignment subject only to the constraint that the sum gives $\mathcal{M}$.

(iv): S-matrix theory (Dyson, 1949; Heisenberg, 1943) defines physical observables exclusively from the asymptotic states $|p_1 \cdots p_n\rangle_{\text{in}}$ and $|k_1 \cdots k_m\rangle_{\text{out}}$. No measurement of any intermediate virtual state is possible. Path Opacity states: two backward trajectories sharing an endpoint are indistinguishable by any metric invariant computed at the endpoint. The endpoint of the backward trajectory is the external state of the S-matrix; the intermediate virtual sub-states are the internal virtual particles.

(v): The optical theorem (Cutkosky, 1960) gives $2\text{Im } \mathcal{M}(p \rightarrow p) = \sum_X \int d\Phi_X |\mathcal{M}(p \rightarrow X)|^2$, which is the constraint that the sum of all transition probabilities equals 1 (unitarity). The Cascade Power formula $\kappa(\Gamma) = 1 - \prod_i (1 - \kappa_i)$ saturates to 1 when $\sum_i \kappa_i = \infty$ (Borel, 1909). In QFT, summing over all virtual contributions (all orders of perturbation theory) gives a unitary result when the perturbation series is well-defined. $\square$ $\square$

7.3 Feynman Diagrams as Labeled Compositions

Proposition 7.2 (Feynman Diagram Count). *The number of topologically distinct Feynman diagrams at perturbative order n in a quantum field theory with d field types (particle species) is bounded below by $T(n, d) = d(d+1)^{n-1}$.*

Proof. A Feynman diagram at order n consists of n vertices, each labeled by one of d interaction terms (field types). The n vertices can be arranged into an ordered sequence of k groups for any composition $(c_1, \dots, c_k)$ of n , and each group can be of any of d types. This is precisely a labeled composition of n in d dimensions, and distinct labeled compositions produce topologically distinct diagrams (different vertex orderings and type assignments yield different loop structures). $\square$ $\square$

Remark 7.3. This provides a combinatorial lower bound on the number of Feynman diagrams. The known growth of QED diagrams as $(2n-1)!! = 1 \cdot 3 \cdot 5 \cdots (2n-1)$ for n -loop contributions exceeds $T(n, 3) = 3 \cdot 4^{n-1}$ only for large n ; the composition-inflation formula provides the combinatorial floor, while additional symmetry factors and loop topologies contribute above this floor.

7.4 Ward–Takahashi Identities as Local–Global Decoupling

The Ward–Takahashi identities (Takahashi, 1957; Ward, 1950) are the statement that in QED, the photon propagator and vertex function satisfy

$$q^\mu \Gamma_\mu(p, p') = S_F^{-1}(p') - S_F^{-1}(p), \quad (7)$$

where Γ_μ is the full vertex and S_F is the full fermion propagator. This identity ensures gauge invariance of the physical amplitude despite the gauge-dependence of individual diagrams.

Theorem 7.4 (Ward–Takahashi as Local–Global Decoupling). *The Ward–Takahashi identity is the gauge-field instantiation of the Local–Global Decoupling Theorem: it states that the global physical amplitude is preserved under arbitrary rerouting of virtual photon momenta (local S-value changes), subject only to 4-momentum conservation (mean-recovery constraint).*

Proof. The Ward–Takahashi identity arises from the fact that the gauge transformation $A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$ changes individual Feynman diagram amplitudes but not the physical S-matrix. Each diagram amplitude (local S-value) is gauge-dependent; their sum (global S-value) is not. The Local–Global Decoupling Theorem states that for any global S-value and any assignment of local S-values to subtasks, the expression realising both simultaneously exists. The gauge parameter ξ plays the role of the free local S-value assignment: varying ξ changes the individual diagram amplitudes without changing the physical observable. The 4-momentum conservation at each vertex is the mean-recovery constraint: the sum

of incoming virtual momenta equals the sum of outgoing, exactly as the mean of the virtual sub-coordinates equals the physical global coordinate. $\square$ $\square$

8 Angular Reformulation of Fundamental Constants

8.1 The Partition Propagation Bound

From the partition structure of bounded phase space, the maximum information propagation rate is finite. For a system with partition depth n and level spacing ℓ_n , the propagation speed satisfies

$$c = \sup_n \ell_n \omega_n < \infty, \quad (8)$$

where ω_n is the oscillation frequency at level n . For one complete oscillation cycle ($\Delta\phi = 2\pi$ rad), the distance traversed is $\Delta x = 2\pi r_0$ at the characteristic radius r_0 . Expressing distance in angular units $\theta = x/r_0$:

Theorem 8.1 (Speed of Light in Categorical Angular Units).

$$c = 2\pi \text{ rad/tick}. \quad (9)$$

Proof. One oscillation cycle traverses $\Delta x = 2\pi r_0$ in time $\Delta t = 1$ tick. The speed $c = \Delta x / \Delta t = 2\pi r_0 / \text{tick}$. In angular units (radians, measuring arc length at r_0), $\Delta\theta = \Delta x / r_0 = 2\pi$, so $c = 2\pi \text{ rad/tick}$. $\square$ $\square$

Theorem 8.2 (Planck's Constant in Categorical Angular Units).

$$\hbar = E_{\text{tick}} / (2\pi), \quad (10)$$

where E_{tick} is the energy carried per oscillation tick.

Proof. From $E = \hbar\omega$ with $\omega = 2\pi / \text{tick}$: $E_{\text{tick}} = \hbar \cdot 2\pi$. Solving: $\hbar = E_{\text{tick}} / (2\pi)$. $\square$ $\square$

Theorem 8.3 (Boltzmann's Constant in Categorical Angular Units).

$$k_B = E_{\text{tick}} / \ln(d+1), \quad (11)$$

where d is the dimensionality of the entropy space and $\ln(d+1)$ is the information per composition step.

Proof. Each additional oscillation tick increases the trajectory count by factor $(d+1)$ (Proposition 3.6), contributing $\ln(d+1)$ natural units of entropy. The thermal energy per degree of freedom is $k_B T$, where T is the tick rate (ticks per second). Equating the energy per unit of composition entropy: $k_B = E_{\text{tick}} / \ln(d+1)$. For $d=3$: $k_B = E_{\text{tick}} / \ln 4$. $\square$ $\square$

Theorem 8.4 (Planck Time in Categorical Angular Units).

$$t_P = \sqrt{\hbar G / c^5} = \frac{\sqrt{E_{\text{tick}} \cdot G}}{(2\pi)^3}.$$

The Planck time has two components: $(2\pi)^3$ from the angular geometry of oscillation and $\sqrt{E_{\text{tick}} \cdot G}$ as the unique irreducible product coupling tick energy to gravitational coupling.

Proof. Substitute Theorems 8.1–8.2: $t_P = \sqrt{E_{\text{tick}} G / (2\pi \cdot (2\pi)^5)} = \sqrt{E_{\text{tick}} G} / (2\pi)^3$. $\square$ $\square$

Corollary 8.5 (Irreducibility of G). *The gravitational constant G is the unique fundamental constant that does not reduce to a geometric factor (2π) , a counting quantity (n, d) , or an energy unit (E_{tick}) in the categorical angular framework. It is the sole external coupling input to the partition framework.*

Table 2: Reduction of fundamental constants to categorical angular units.

Constant	SI expression	Categorical form
c	$2.998 \times 10^8 \text{ m/s}$	$2\pi \text{ rad/tick}$
$\hbar$	$1.055 \times 10^{-34} \text{ J} \cdot \text{s}$	$E_{\text{tick}} / (2\pi)$
k_B	$1.381 \times 10^{-23} \text{ J/K}$	$E_{\text{tick}} / \ln(d+1)$
m_0	$\hbar\omega_0 / c^2$	$E_{\text{tick}}\nu_0 / (4\pi^2)$
$E_0 = m_0 c^2$	$m_0 c^2$	$E_{\text{tick}}\nu_0$
t_P	$\sqrt{\hbar G / c^5}$	$\sqrt{E_{\text{tick}} G} / (2\pi)^3$
G	$6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$	irreducible

9 Discussion

9.1 The Current LHC Trigger in Composition-Inflation Terms

The ATLAS Level-1 trigger has a fixed latency of $2.5 \mu\text{s} = 100$ bunch crossings (ATLAS Collaboration, 2020). In composition-inflation terms, this corresponds to depth $n = 100$ in $d = 3$ S-entropy dimensions, giving a trigger pattern space of $T(100, 3) = 3 \cdot 4^{99} \approx 3.0 \times 10^{59}$. The Planck depth is $n_P = 60$; the trigger operates at $n/n_P = 1.67$, well above the Planck depth.

The current ATLAS trigger menu contains approximately 10^3 distinct trigger paths (ATLAS Collaboration, 2020), corresponding to depth $n \approx \log_4(10^3/3) + 1 = 6$ — only 10% of the Planck depth. The composition-inflation framework therefore predicts that $\approx 90\%$ of the available categorical resolution in the L1 timing is unused by the current trigger specification. This suggests substantial room for improvement in trigger algorithms that exploit composition structure rather than flat coincidence conditions.

9.2 Cascade Catalysis and Renormalization

Theorem 9.1 (Renormalization as Cascade Saturation). *The Wilsonian renormalization group flow (Callan, 1970; Symanzik, 1970; Wilson, 1971) corresponds to cascade saturation: the running coupling constant $g(\mu)$ approaches a fixed point iff the sum of catalytic powers of the virtual loop corrections diverges.*

Proof. The Callan–Symanzik equation $\mu \partial g / \partial \mu = \beta(g)$ describes how the coupling g changes with the renormalization scale μ . A fixed point g^* satisfies $\beta(g^*) = 0$.

In the cascade picture, each loop correction at order n contributes a catalytic power $\kappa_n = |\beta_n|g^{n-1}$ (proportional to the loop-correction amplitude). The total catalytic power of the cascade of loop corrections from scale μ_0 to μ is $\kappa_{\text{total}} = 1 - \prod_n (1 - \kappa_n)$. The cascade saturates to power 1 (i.e. $g(\mu)$ reaches g^*) iff $\sum_n \kappa_n = \infty$, which by the Borel–Cantelli analogy (Borel, 1909) holds iff the β -function series does not decay faster than $1/n$. For asymptotically free theories ($g^* = 0$, $\beta(g) = -\beta_0 g^2 + \dots$), the sum $\sum_n \beta_n g^{n-1}$ diverges as $g \rightarrow 0$ from above, confirming saturation to the trivial fixed point. $\square$ $\square$

9.3 Dark Matter and Virtual-State Suppression

Theorem 9.2 (Dark Sector as Virtual-State-Suppressed). *A particle sector in which backward trajectory completion is restricted to physical sub-states (Theorem 6.5) has $\Theta(N)$ interaction complexity and consequently cannot reach thermal equilibrium with the visible sector on any finite timescale proportional to $\log N$.*

Proof. If the dark sector is constrained to physical intermediate states ($\mathbf{S} \in [0, 1]^3$ at every step), Theorem 6.5 gives interaction complexity $\Theta(N)$. The visible sector, using virtual sub-states, has interaction complexity $\mathcal{O}(\log N)$. The ratio of dark-to-visible interaction rates is $\Theta(N)/\mathcal{O}(\log N) = \Theta(N/\log N) \rightarrow \infty$. For the cosmological particle count $N \sim 10^{80}$, the dark sector interaction rate is suppressed by $10^{80}/\log(10^{80}) \approx 10^{78}$ relative to the visible sector. Gravitational coupling, which depends only on G (the unique irreducible constant), operates at the same rate for both sectors, providing the only dark–visible coupling. $\square$ $\square$

Remark 9.3. Theorem 9.2 does not constitute a dark matter model but identifies the categorical structure that would produce the observed phenomenology: invisible to electromagnetic interactions (no off-shell photon exchange without virtual sub-states), gravitationally coupled (through G , the irreducible constant), and thermally decoupled from the visible sector.

10 Conclusion

We have established a unified mathematical framework connecting the real-time event selection problem of LHC trigger systems to categorical mechanics, composition-inflation, and S-entropy trajectory completion. The framework produces six principal results:

(1) **Composition-Inflation (Theorem 3.4):** A trigger with d subsystems and n bunch crossings can discriminate $T(n, d) = d(d+1)^{n-1}$ event patterns—exponential, not linear, in n .

(2) **LHC Planck Depth (Theorem 4.3):** $n_P = 60$ for the LHC at $d = 3$; the L1 trigger latency of 100 bunch crossings operates at 1.67 times the Planck depth, placing it in the super-Planck categorical resolution regime.

(3) **Structural Incorruptibility (Theorem 5.4):** Timing-only triggers have no injection surface; security is topological, not cryptographic.

(4) **Virtual Particle Identification (Theorem 7.1):** Virtual particles in QFT are formal virtual sub-states; their five properties (off-shell, zero net work, path opacity, catalytic power, necessity) are categorical theorems.

(5) **Constant Reduction (Table 2):** Six of seven fundamental constants reduce to tick counts and 2π geometry; G is the unique irreducible coupling.

(6) **HL-LHC Advantage (Theorem 4.5):** The precision timing upgrade ($d : 3 \rightarrow 4$) reduces n_P from 60 to 52, recovering 200 ns of categorical resolution headroom.

The framework provides formal trigger design principles grounded in first-principles physics: the optimal trigger depth is n_P , determined by the bunch clock frequency and the number of independent subsystems; adding subsystems reduces n_P logarithmically; and the trigger menu is bounded not by engineering constraints alone but by the composition-inflation state space $T(n, d)$.

Acknowledgements

This work was supported by the AIME Registry for Artificial Intelligence.

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